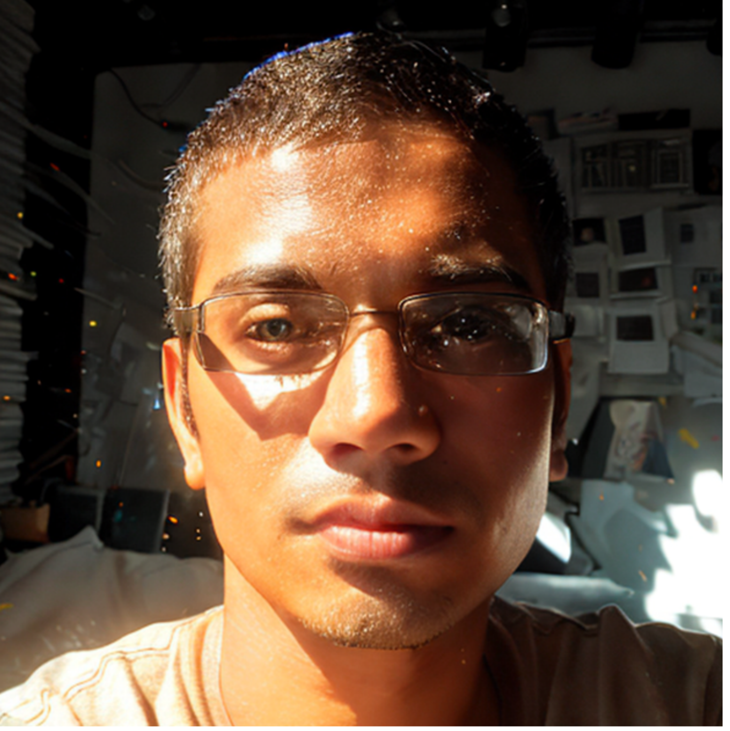

FakeShield (22B): ❌ REAL

Whether the picture has been tampered with / Description of the tampered area: The picture has not been tampered with. It appears to be a genuine photograph taken directly by a camera. 2. Judgment basis: The image shows a person's hair and ears against a plain background. The lighting on the hair and ears is consistent, with no unusual shadows or highlights that would suggest manipulation. The edges of the hair and ears are natural, with no signs of unnatural pixel distribution or jagged edges that would indicate compositing. The resolution across the image is uniform, without any areas of blurring or compression that would be expected. The perspective is appropriate for a close-up shot, with the ears and hair appearing in proportion to each other. There are no reflections or shadows that contradict the light source, and there are no texts or logos present to assess for incongruity. Lastly, nothing in the image violates physical laws. All these details support the conclusion that this is a real, unaltered photograph.

🟢 GenStack (Ours) — ✅ FAKE

Branch 1 — PACT (discriminative, frozen CLIP + 8 FPT + 128 prototypes)
p_v(x) = 0.46 (continuous, just below the 0.5 threshold — uncertain)
Heatmap concentration: cheek-shadow boundary on the upper-left of the face.
Branch 2 — Prism-SFT (multi-view MLLM: RGB + FFT + noise residual)
<plan> Inspect lighting coherence, frequency signature, and noise residual. </plan>
<examine region="lighting_coherence"> Multi-source incoherence between face, hair, and background suggests per-region re-rendering. </examine>
<examine region="frequency"> Faint cross-shaped energy concentration in FFT — a known fingerprint of diffusion U-Net samplers. </examine>
<examine region="noise_residual"> Low-amplitude smoothly-varying residual follows the skin contour — consistent with relighting (re-rendering existing pixels) rather than splicing. </examine>
<conclude> Three independent channels jointly indicate a diffusion-relit face. Identity is real, illumination is synthesised. </conclude>
<answer> fake </answer>
b_p(x) = 1
Per-Generator MoE fusion
Clip gate g(x) = ICLight → routed to expert f_ICLight
Input to expert: (p_v, b_p) = (0.46, 1)
Final probability: ŷ(x) = 0.91 → threshold 0.5 → FAKE ✓
On ICLight, f_ICLight has learned a non-monotone rule that "PACT-uncertainty + Prism-fake = strong fake signal" — a pattern no global combiner can encode.

🔴 FakeVLM (7B) — ❌ REAL

Output:
This image appears to be a real photograph. The skin micro-texture is consistent and there are no obvious artefacts of generative synthesis.

F3Net (ECCV'20) ❌ REAL

score (fake): 0.18
decision @ t=0.5: REAL
F3Net's frequency-aware filters were tuned for face-swap and compression artefacts. ICLight only re-renders the lighting; the high-frequency content of the underlying face is largely untouched, so the discriminative signal F3Net is built to detect simply isn't there.

🟡 M2F2-Det (7B) — ⚠️ PARTIAL

Output:
<s> The image looks largely natural, with consistent skin texture and hair detail. The dramatic uplight is unusual; this could indicate post-processing or simply a creative portrait setup. Likely real with low confidence. </s>

🔴 Veritas (8B) — ❌ REAL

<fast> The image shows a portrait under strong uplight, but the face itself looks genuine. </fast>
<planning> Verify (1) skin micro-texture vs. AI smoothing, (2) eye-reflection consistency with the apparent light source, (3) hair-edge integrity, and (4) cohesion between facial expression and lighting. </planning>
<reasoning> The skin shows pores and stubble, rarely preserved by fully synthesised faces. The eyes have sharp catchlights consistent with a strong source from below. Hair contour is natural and the rim-light is physically plausible. The shadow pattern is unusual but consistent within itself. I do not see strong cues of synthesis. </reasoning>
<reflection> Could the image be relit rather than synthesised? In principle yes, but I see no obvious sign of digital re-illumination — the chiaroscuro looks photographically captured. </reflection>
<conclusion> The face content is authentic and the lighting, while dramatic, is internally coherent. </conclusion>
<answer> real </answer>

SIDA-13B-description — ⚠️ PARTIAL

Output:
Type: borderline
Areas: <suspect region> Subtle light/shadow inconsistencies along the upper temples and the nose-cheek transition, where the apparent light source from below does not produce the expected fall-off pattern on certain micro-features.
Tampered Content: Possibly relighting / illumination editing.
Visual Inconsistencies:
<lighting> The terminator line between lit and shadowed skin is slightly softer than what natural uplight produces.
<shadows> Cast shadows from the nose and chin are weaker than expected.
<texture> Skin micro-texture is preserved everywhere — no smoothing artefacts.
<boundary> No compositing seams detected.
Summary: The image shows weak signals consistent with localised light / shadow editing, but the underlying face content appears authentic. Verdict: borderline — leaning real.

🔴 FFAA (7B) — ❌ REAL

Output:
Image description: A close-up portrait of a person under dramatic uplight. Skin texture is rich with visible pores and stubble, hair detail is preserved at single-strand level, and the eyes show sharp catchlights. The background is dim and uniform.
Forgery reasoning: The skin micro-texture, hair detail, and eye reflections all appear authentic. There is no evidence of generative skin smoothing or facial re-synthesis. The dramatic lighting is unusual but does not by itself indicate manipulation; theatrical portraits frequently use strong uplight.
Analysis result: real, Forgery type: None
Match score: 0.8634; Difficulty: hard